

Milestone 4 Group Report

Team: TFC

Tianhao Cao, Yusen Huang, Marco Wang, Darwin Zhang

1 Methodology

1.1 Method 1: Neighborhood Attack for VLMs

1.1.1 Model Description

This method adapts the neighborhood attack paradigm to the multimodal setting by exploiting both the finetuned VLM and the original base model (SmolVLM-256M-Instruct before finetuning). The core intuition is that a model that has memorized a training sample will assign a distinctly lower loss to the exact (image, text) pair than to semantically close but slightly perturbed neighbors. Non-members, having never been seen, show no such "local minimum" in loss around the original sample.

For each sample, we extract a 17-dimensional feature vector across four groups:

- **Base losses (4 features):** The raw cross-entropy loss from the finetuned model (`loss_ft`) and the base model (`loss_base`), along with their difference (`ref_diff = loss_base - loss_ft`) and ratio (`ref_ratio`). The reference model comparison is key: if finetuning caused memorization, `loss_ft` should drop significantly relative to `loss_base` for members only.
- **Text neighborhood features (4 features):** We generate 5 text neighbors by randomly dropping 10% of words from the assistant response (keeping the user question intact and the image fixed). The **neighborhood gap** (`text_nb_gap_ft`) measures how much the loss increases when the text is perturbed — a large gap indicates the model has strongly memorized the specific wording of the original answer. Analogous gaps are computed for the base model, and a **relative gap** (`text_nb_relative_gap = text_nb_gap_ft - text_nb_gap_base`) isolates the finetuning-specific memorization from pre-existing language preferences.
- **Image neighborhood features (4 features):** We generate 5 image neighbors by cycling through three perturbation strategies: (1) additive Gaussian noise ($\sigma = 25$), (2) MAE-style patch masking (masking 15% of 32×32 patches with the mean color), and (3) random crop-and-resize (retaining 70–90% of the image). The text is kept fixed. Analogously to text, `img_nb_gap_ft` measures sensitivity to image perturbation, and `img_nb_relative_gap` controls for the base model’s visual sensitivity.

Method	AUC	TPR@FPR=0.1
Baseline	0.50082	0.12
Neighborhood Attack (VLM)	0.5988	0.1867

Table 1: Membership inference attack results on the SmolVLM model.

- **Cross-modal binding features (3 features):** If the model has memorized the exact (image, text) *pair*, both modalities should simultaneously contribute to a low loss. We capture this with `cross_modal_gap = text_nb_gap_ft - img_nb_gap_ft` (asymmetry between modalities), `binding_score = text_nb_gap_ft × img_nb_gap_ft` (joint sensitivity, nonzero only when both gaps are large), and `total_nb_gap` (their sum).
- **Text statistics (2 features):** Word count (`text_len`) and vocabulary diversity (`unique_ratio`), included as control variables since longer or more repetitive texts may systematically affect loss regardless of membership.

All features are standardized using `StandardScaler` before being fed into an XGBoost classifier (with fallback to `GradientBoostingClassifier` if XGBoost is unavailable), trained with 300 estimators, maximum depth 4, and learning rate 0.05.

1.1.2 Difference from Baseline

The baseline uses the finetuned model’s loss as a single scalar signal. This method introduces three fundamental extensions. First, the **reference model comparison** distinguishes membership from naturally easy samples: a low loss on its own may simply indicate the sample is linguistically simple; a low loss *relative to the base model* more specifically indicates finetuning-induced memorization. Second, the **neighborhood gap** measures local loss curvature rather than absolute loss — members sit at sharper local minima in both text and image input spaces. Third, the **cross-modal binding** features are unique to the VLM setting and test whether memorization is tied to the specific pairing of an image and its caption, rather than either modality alone.

1.1.3 Results & Discussions

The neighborhood attack achieves an AUC of 0.5988 and TPR@FPR=0.1 of 0.1867, representing a meaningful improvement over both the baseline (AUC 0.500) and the Min-K% Prob method (AUC 0.513). The 0.7500 validation accuracy further suggests the classifier is learning a genuine signal rather than random noise. However, the AUC remains well below 0.7, indicating that the method captures only a partial memorization signal.

The reasons why this model does not achieve stronger results:

Perturbation granularity. Text neighbors are generated by dropping whole words at random, which is a coarse perturbation. If the model’s memorization is tied to specific sub-word tokens or syntactic structures rather than word-level patterns, word-drop neighbors may not reliably escape the memorized region in input space, underestimating the neighborhood gap for members.

Image perturbation fidelity. The three image perturbation strategies (Gaussian noise, patch masking, random crop) alter low-level pixel statistics but may preserve high-level semantic content. A VLM that has memorized a training image may still assign low loss to a semantically similar perturbed version, compressing the gap between members and non-members in the image neighborhood features.

Base model alignment. The reference model comparison (`ref_diff`, `ref_ratio`) assumes the base model provides a clean counterfactual. In practice, the base model (SmolVLM-256M-Instruct) has itself been instruction-tuned, so some samples may already be partially familiar to it, weakening the contrast with the finetuned model.

Dataset characteristics. With 75% accuracy at a likely balanced threshold, the classifier may be largely exploiting distributional differences in text length or vocabulary richness (`text_len`, `unique_ratio`) rather than true memorization signals, which would explain why AUC — a threshold-independent metric — lags behind accuracy.

Cons:

- Generating 5 text neighbors and 5 image neighbors per sample requires 12 forward passes per sample (original + 10 neighbors, each through both models), making this method approximately 24× more expensive than the baseline.
- Word-drop perturbation does not preserve grammaticality, potentially producing out-of-distribution inputs that confuse the model independently of membership status.
- XGBoost with 17 features on a potentially small labeled training set risks overfitting, especially given the noisy neighborhood signals.
- Cross-modal binding features (`binding_score`, `total_nb_gap`) are products or sums of already noisy gap estimates, which may amplify variance rather than add information.

Possible Improvements to this method:

- Replace word-drop with paraphrase-based text neighbors (e.g., using a masked language model to substitute tokens), which would produce more naturalistic perturbations closer to the original in semantic space.
- Use semantics-preserving image augmentations (e.g., color jitter, small rotations) instead of destructive perturbations, so that the neighborhood boundary is tighter around the original image.
- Increase the number of neighbors and aggregate their statistics more robustly (e.g., median instead of mean) to reduce the variance of gap estimates.
- Incorporate internal activation signals (e.g., hidden state norms in cross-attention layers) in addition to output-level loss, as memorization in VLMs may be encoded in fusion layers invisible to output probing.

1.2 Method 2: Multi-Feature MIA with Min-K% Prob

1.2.1 Model Description

This method extends the baseline by moving beyond aggregate loss to extract per-token log probabilities from the SmolVLM model. For each sample, we feed the image-text pair through the model and compute token-level log probabilities across the entire output sequence. From these token-level signals, we derive seven features:

- **Loss:** Mean per-token cross-entropy loss (retained from baseline for continuity).
- **Perplexity:** $\exp(\text{loss})$, a nonlinear transform that amplifies differences at high loss values, making it easier for the classifier to separate borderline cases.
- **Min-K% Prob:** The average log-probability of the bottom 20% lowest-probability tokens. This is the core feature. When a model has memorized a training sample, it assigns reasonable probability even to "surprising" or rare tokens in that text. For non-members, those hard tokens remain genuinely difficult. This creates a wider gap in Min-K% Prob between members and non-members than what aggregate loss captures. This method is based on the Min-K% Prob attack proposed by Shi et al. (2024).
- **Max token loss:** The single worst token's cross-entropy loss. Members should exhibit lower values because memorization reduces extreme prediction failures.
- **Std of token losses:** Standard deviation across all token losses. Members are expected to have more uniform losses (lower variance) since the model has learned the full sequence consistently.
- **Zlib ratio:** Loss divided by the zlib-compressed byte length of the text. This normalizes by intrinsic text complexity — simple, repetitive texts naturally produce low loss regardless of membership, so dividing by compressibility controls for this confound.
- **Num tokens:** The number of target tokens in the sequence, included as a control variable since sequence length may systematically affect other features.

All features are standardized using `StandardScaler` before being fed into a Logistic Regression classifier with balanced class weights. The baseline treats the model as a black box producing a single loss scalar per sample. This method opens up the token-level distribution and extracts multiple complementary signals. The key conceptual difference is that Min-K% Prob focuses specifically on the **hardest tokens** — the tail of the distribution — rather than the average, which is a fundamentally different measurement of memorization. The zlib ratio additionally controls for text complexity, which the baseline does not account for.

1.2.2 Results & Discussions

All individual features achieved AUC values near 0.50, and the combined model only reached 0.51272. This indicates that standard text-based MIA signals — including Min-K% Prob, which has shown strong results on text-only LLMs — are not sufficiently discriminative for this vision-language model. The reasons why this model does not have good results:

Method	AUC	TPR@FPR=0.1
Baseline	0.50082	0.12
Multi-Feature Min K Probability	0.51272	0.08

Table 2: Caption

Multimodal information distribution. In a VLM, memorization signals may be distributed across both the visual and textual modalities. Our features only capture text-side token probabilities. If the model primarily memorizes visual features or specific image-text alignments rather than the text itself, token-level text signals will be weak.

Model architecture. SmolVLM processes images through a vision encoder before fusing with text tokens. Memorization may manifest in the cross-attention or fusion layers rather than in the output token probabilities, making it invisible to output-only probing methods.

Cons:

- All features are text-side only; no image-side memorization signals are captured.
- Softmax-based features from the model’s output layer may not reflect internal memorization patterns in earlier layers.
- Logistic Regression assumes linear separability, which may not hold for these features.

Possible Improvement to this method: Experiment with $K = 5\%, 10\%, 30\%, 50\%$ for Min-K% Prob, and restrict computation to only the non-top-k tokens (which are the minimum probability tokens) of the visual tokens or the join image-text likelihood, rather than processing the full token distribution, to better capture memorization signals specific to VLMs.

1.3 Method 3: Lightweight Single-Model MIA via Complexity Calibration

1.3.1 Model Description

To address the prohibitive computational costs associated with multi-model reference attacks (which require passing data through both a fine-tuned target model and a baseline reference model), this method presents a highly optimized, single-model approach. It relies exclusively on the fine-tuned VLM, significantly reducing inference overhead. To compensate for the lack of a reference model for calibration, we leverage intrinsic text complexity and statistical token distribution metrics.

For each image-text pair, we extract token-level log probabilities from the fine-tuned SmolVLM and compute the following features:

- **Loss:** Mean per-token cross-entropy loss.
- **Perplexity:** $\exp(loss)$, which amplifies differences in high-loss regions.

- **Min-K% Prob:** The average log-probability of the bottom 20% lowest-probability tokens (Shi et al., 2024), effectively isolating the hardest tokens to detect memorization signatures without sequence-level smoothing.
- **Max token loss:** The single worst token’s cross-entropy loss, where members typically exhibit lower extremes.
- **Std of token losses:** Standard deviation across all token losses to capture sequence-level prediction consistency.
- **Zlib ratio:** The average loss divided by the zlib-compressed byte length of the target text. Because we are not using a base model to calibrate easy versus hard text, we use algorithmic compression as a proxy. Simple, repetitive texts yield high compressibility and naturally lower losses; dividing by the zlib byte length normalizes the loss, preventing inherently simple text from being falsely flagged as memorized data.
- **Num tokens:** The total sequence length as a control variable.

All features are standardized using **StandardScaler**. However, unlike Method 2 which utilized Logistic Regression, these features are fed into an **XGBoost classifier** (a tree-based gradient boosting model).

The key conceptual difference between Method 3 and previous methods is the integration of the Zlib ratio as a standalone calibration tool combined with a nonlinear classifier. While Logistic Regression assumes linear separability, XGBoost is capable of learning complex, non-linear interactions between statistical token features.

1.3.2 Expected Results & Discussions

Method	AUC	TPR@FPR=0.1
Baseline	0.50082	0.12
Multi-Feature Min K Probability	0.51272	0.08
Lightweight Complexity Calibration	0.56050	0.18

Table 3: Anticipated Membership Inference Attack performance across tested methods.

Based on the inclusion of the Zlib compression proxy, we anticipate this method to achieve an AUC of approximately 0.585 with a TPR of 0.18 at a 0.1 FPR. The primary driver of this expected improvement is XGBoost’s ability to map non-linear interactions, such as sequences with high Perplexity but exceptionally low Zlib ratios, which linear baselines fail to separate. By relying on intrinsic text complexity rather than an external neural reference model, we bypass the need for expensive multi-model inference while still gaining a functional calibration signal.

Cons:

- **Modality Blindness:** Like Method 2, this approach relies entirely on text-side token probabilities. It remains blind to purely visual memorization or cross-modal binding parameters.

- **Compression Proxy Limits:** Zlib compression is a syntactic algorithm. While it effectively measures character-level repetition, it does not perfectly align with semantic likelihood as perceived by a VLM.
- **Overfitting Risk:** XGBoost models can quickly overfit on small, subsetted feature dimensions, requiring careful hyperparameter tuning.

Future Improvements and Model Scaling: Testing lightweight models is a strategic and necessary step in our pipeline. If a computationally efficient model can generalize similarly to state-of-the-art multi-model frameworks, it serves as a highly practical baseline. While advanced methods are occasionally required, scaling complexity is only justified if empirical performance and supporting literature dictate the need. Moving forward, a promising option to improve upon this lightweight foundation is to combine the Min-K% Probability approach with visual ablation. By adding a feature that computes the loss difference when the real image is replaced with a blank image, we can directly measure cross-modal memorization. This layered approach ensures we maintain efficiency while incrementally adding targeted, literature-backed complexity to capture missing multi-modal signals.

2 Literature Review

Paper 1: M⁴I: Multi-modal Models Membership Inference

This paper addresses the privacy risks of multi-modal machine learning models by proposing the first membership inference attack (MIA) framework specifically designed for models that process multiple modalities. Prior MIA research had focused almost exclusively on single-modal models such as image classifiers or tabular data models, leaving the rapidly growing field of vision-language models largely unexplored. The authors argue that the cross-modal nature of these systems—where an image input produces a text output, for example—introduces unique vulnerabilities that require new attack strategies beyond simply probing a single output distribution.

Short summary: Main Task: Determining whether a specific data record (such as an image-caption pair) was used to train a multi-modal model. This is formulated as a binary classification problem: given a target sample and query access to the model, classify it as a member (seen during training) or a non-member.

Models: The paper proposes two attack methods—**Metric-based M⁴I (MB-M⁴I)**, which uses text generation similarity metrics (**ROUGE** and **BLEU**) between the model’s output and the target caption to infer membership; and **Feature-based M⁴I (FB-M⁴I)**, which employs a pre-trained shadow multi-modal feature extractor and uses **Euclidean distance** in the shared feature space to compare input-output similarity. Target captioning models follow the **Show-and-Tell** setting—encoder-decoder architectures with a **ResNet-152** or **VGG-16** image encoder and an **LSTM** text decoder.

Main Contribution: M⁴I presents itself as the first work to propose an MIA framework explicitly targeting multi-modal models. The metric-based attack achieves an average success rate of **72.5%**, while the feature-based attack achieves **94.83%** in unrestricted scenarios.

These results demonstrate that membership inference attacks are effective against multi-modal models and establish a foundational baseline for subsequent VLM privacy research.

Citation: Hu, P., Wang, Z., Sun, R., Wang, H., & Xue, M. (2022). M^AI: Multi-modal Models Membership Inference. Advances in Neural Information Processing Systems, 35 (NeurIPS 2022). arXiv:2209.06997.

Paper 2: OpenLVLM-MIA: A Controlled Benchmark Revealing the Limits of Membership Inference Attacks on Large Vision-Language Models

This paper critically examines the reliability of existing MIA research on large vision-language models (LVLMs), raising an important methodological concern: many high attack success rates reported in prior work may be artifacts of **distributional bias** in dataset construction rather than evidence of genuine memorization detection. When member and non-member samples come from different distributions (e.g., different time periods, sources, or collection methods), an attack may simply learn to distinguish those distributions rather than identify true training membership. To address this, the authors introduce a carefully controlled benchmark designed to remove such biases.

Short summary: Main Task: Benchmarking the effectiveness of state-of-the-art MIA methods on LVLMs under fair, unbiased evaluation conditions. The paper asks whether existing MIA methods can truly detect training membership, or whether prior high success rates are inflated by distributional shortcuts.

Dataset: The authors introduce **OpenLVLM-MIA**, a controlled benchmark of **6,000 images** (1,000 members and 1,000 non-members per training stage), with ground-truth membership labels across three distinct LVLM training stages: (1) vision encoder pre-training, (2) projector pre-training, and (3) instruction tuning. The member and non-member distributions are carefully balanced to remove any exploitable data bias.

Models: The benchmark is built on an **OpenCLIP + LLaVA** architecture with a fully transparent, open-source training pipeline. The evaluation uses a **gray-box setting**, where the attacker has access to the model’s output logits and generated text (but not internal weights). Evaluated MIA methods include **Perplexity**, **Min-K% Probability**, and **Max Rényi** variants.

Main Contribution: The paper demonstrates that when distributional bias is eliminated, SOTA MIA methods converge to **near-random-chance performance** on LVLMs. This suggests that much of the progress reported in prior VLM-MIA literature reflects dataset construction artifacts rather than genuine privacy vulnerabilities. OpenLVLM-MIA provides a rigorous and transparent foundation for future MIA research on LVLMs, and directly motivates the need for more sophisticated attack strategies beyond simple likelihood-based scoring.

Citation: Miyamoto, R., Fan, X., Kido, F., Matsumoto, T., & Yamana, H. (2025). OpenLVLM-MIA: A Controlled Benchmark Revealing the Limits of Membership Inference Attacks on Large Vision-Language Models. arXiv:2510.16295.

3 Contributions

3.1 Milestone 5

Member	Contributions	Percentage
Tianhao Cao	Methods	25%
Yusen Huang	Methods	25%
Marco Wang	Literature	25%
Darwin Zhang	Methods	25%

Yusen Huang’s contribution was completing the Method 2: Multi-Feature MIA with Min-K% Prob and the percentage is 25% of the total. **Total:** 100%